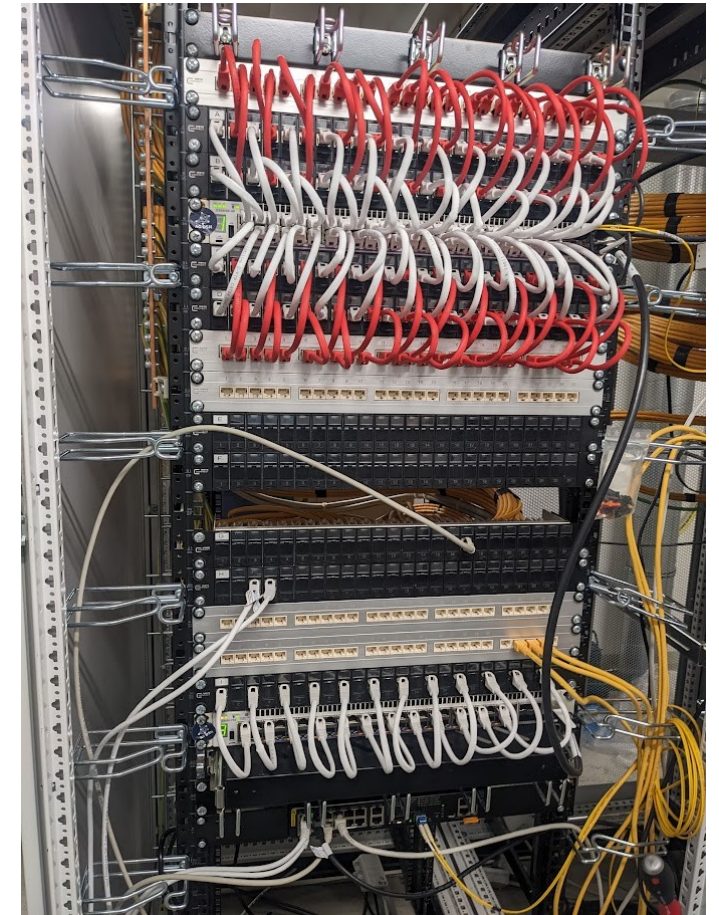


Alexander Lindner & Jonathan Beyer

AG DSN

Who we are?

- founded in 1996
- part of the StuRa of TU Dresden
- ~ 60 volunteers
- 4800 members
- 4 offices



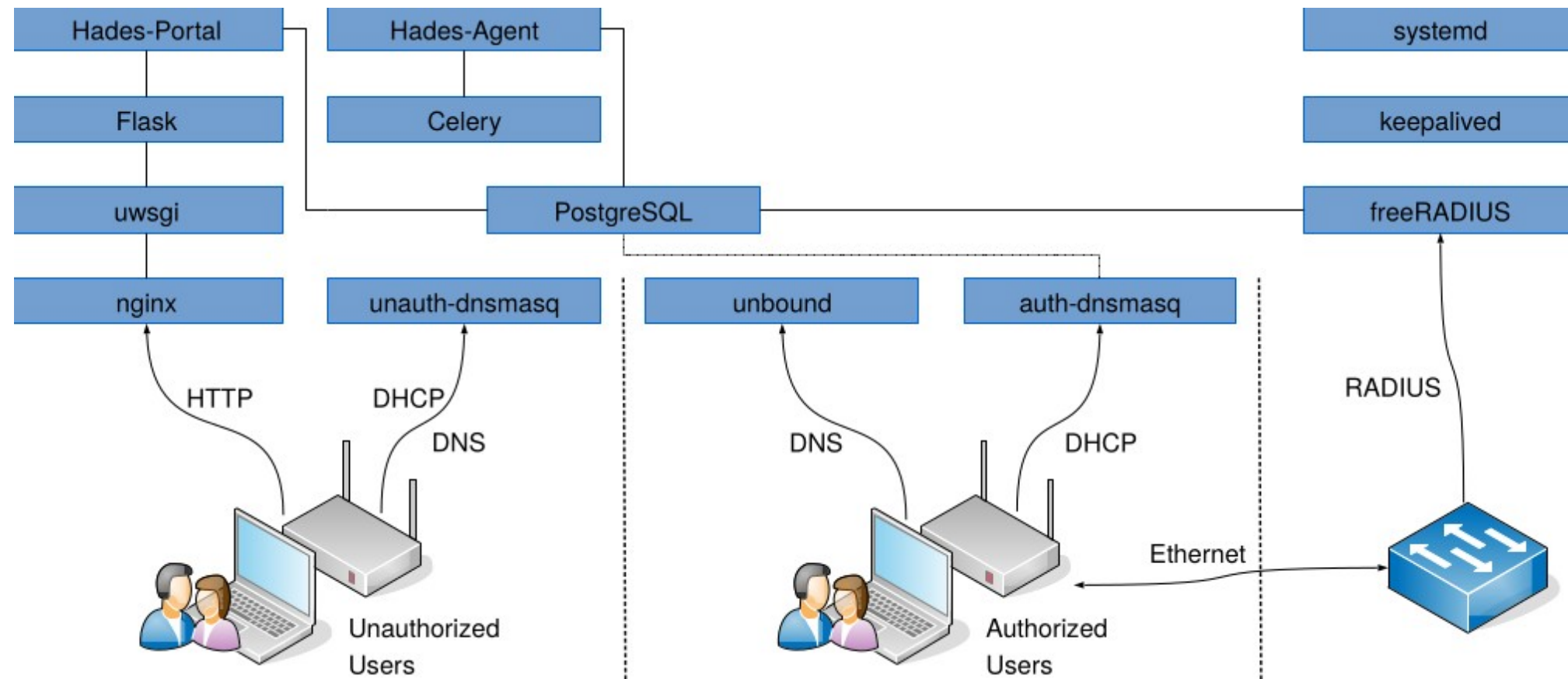


network uplink

- 2x 10G uplink
- 20.000 ipv4 addresses
- Ipv6 /36 subnet



Hades – user authorization



Wifi setup

- SSID's „agdsn“, „agdsn-mpsk“, „eduroam“, „igh“
- 4 wifi controllers
- 667 access points
- 2 nat gateways
- WPA2/3-Enterprise



Plans: Wifi, IPv6 & vlan per user

- every user should become their own network
- we operate NAT & firewall protection
- private network is available at every access point and at wall port in the room



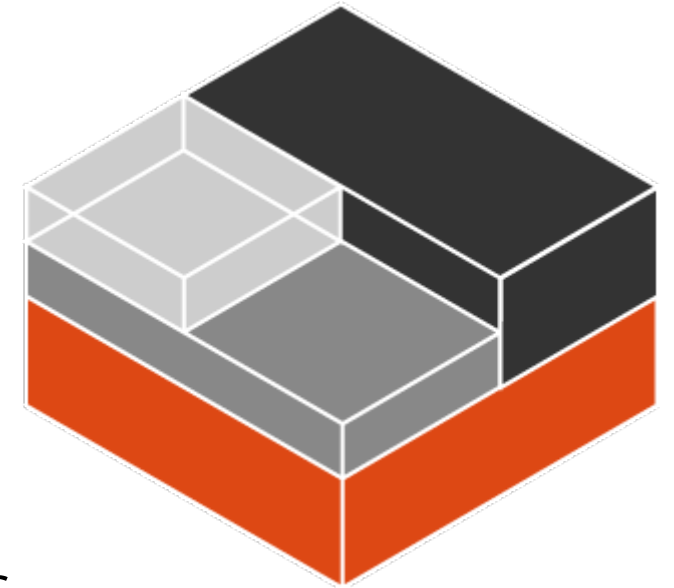
routers: ALE OmniSwitch 6860

- 200G stacking
- industry leading 95W Power over Ethernet
- vrf's to seperate user- & managment-network
- QoS, ACLs, Layer-2/Layer-3 switching, VLAN stacking and even IPv6
- Supports SPB-M for bridging and routed service



virtualization

- oVirt Cluster with ~120 virtual machines
- Red Hat stop maintain oVirt from February 2022, from this point on it's a community project
- from this point on we test Xen, ProxMox, OpenStack and Incus
- Incus is a container and virtual-machine manager



lessons learned

- redundancy protects against pain
- never update two servers BIOSes at same time, one broken server is enough
- limit of default vlans $4096 < \text{our member count}$, we need special protocols e.g. spbm

Test Setup

Distribution Switch



Access Switch



More Protocols

- Bidirectional Forwarding Protocol (BFD)
- Shortest Path Bridging (SPB)
- Intermediate System to Intermediate System (IS-IS)
- Remote Authentication Dial-In User Service (RADIUS)
- RADIUS Sec

We want you!

- You enjoyed the presentation?
- Want to learn more about networking?
- Get to know new people sharing the same interests?

Visit us!



Questions?



Dein Studentennetz in Dresden

Ehrenamtliches Wohnheimnetzwerk seit 1996



<https://agdsn.de>



@ag_dsn



du-bist-gefragt@agdsn.de

